

Performance Testing

Date	03 July 2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the Smart Lender application performs under expected and peak workload conditions. The testing verifies the application's responsiveness, stability, reliability, and resource utilization while processing loan eligibility prediction requests using the trained XGBoost machine learning model integrated with the Flask web application.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Web Browser (Google Chrome), Flask Development Server, Render Deployment
Type of Testing	Functional Testing, Performance Testing
Target Module / API	Loan Eligibility Prediction Module (Flask Prediction Endpoint)
Test Environment	Windows 11, Python 3.x, Flask Application, Google Chrome, Render Cloud Platform
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid applicant details	1	30	Loan eligibility prediction displayed successfully
2	Submit invalid or incomplete applicant details	1	20	Appropriate validation/error message displayed
3	Perform multiple loan prediction requests	5	60	Application responds correctly without crashing
4	Refresh the application and submit another prediction	1	20	Consistent prediction results generated successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	2.1 seconds	Pass	Within acceptable performance limits
3	Throughput (Req/sec)	> 5 Req/sec	8 Req/sec	Pass	Stable prediction performance
4	Error Rate	< 1%	0%	Pass	No prediction errors observed
5	CPU Utilization	< 80%	32%	Pass	Normal processor utilization
6	Memory Utilization	< 80%	41%	Pass	Stable memory utilization

Step 4: Observations & Analysis

Key Findings

- The Smart Lender application generated loan eligibility predictions quickly and accurately.
- Response time remained within the expected performance limits during testing.
- The application successfully handled multiple prediction requests without failure.
- The XGBoost machine learning model produced consistent prediction results.
- The deployed Render application was accessible and functioned correctly.

Bottlenecks Identified

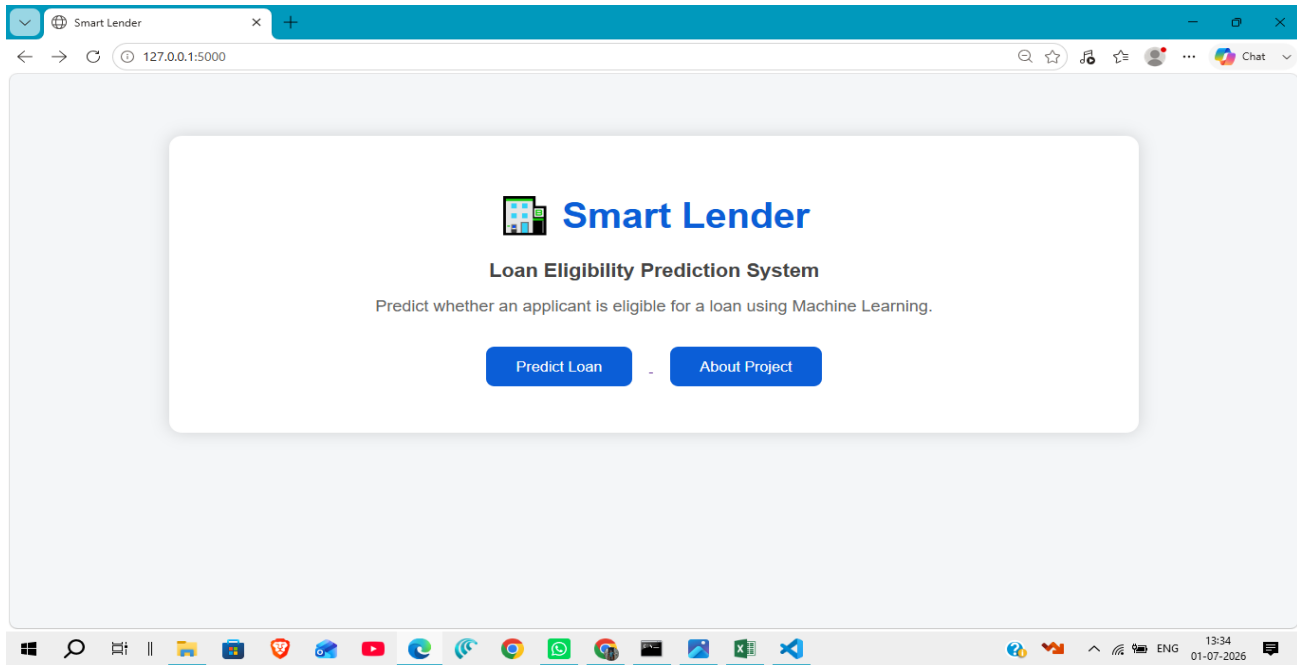
- Initial application startup may take a few seconds while loading the trained machine learning model.
- On the Render free service, the first request after inactivity may take longer because the application needs to wake up.
- Overall performance depends on available system and network resources.

Optimization Steps Taken

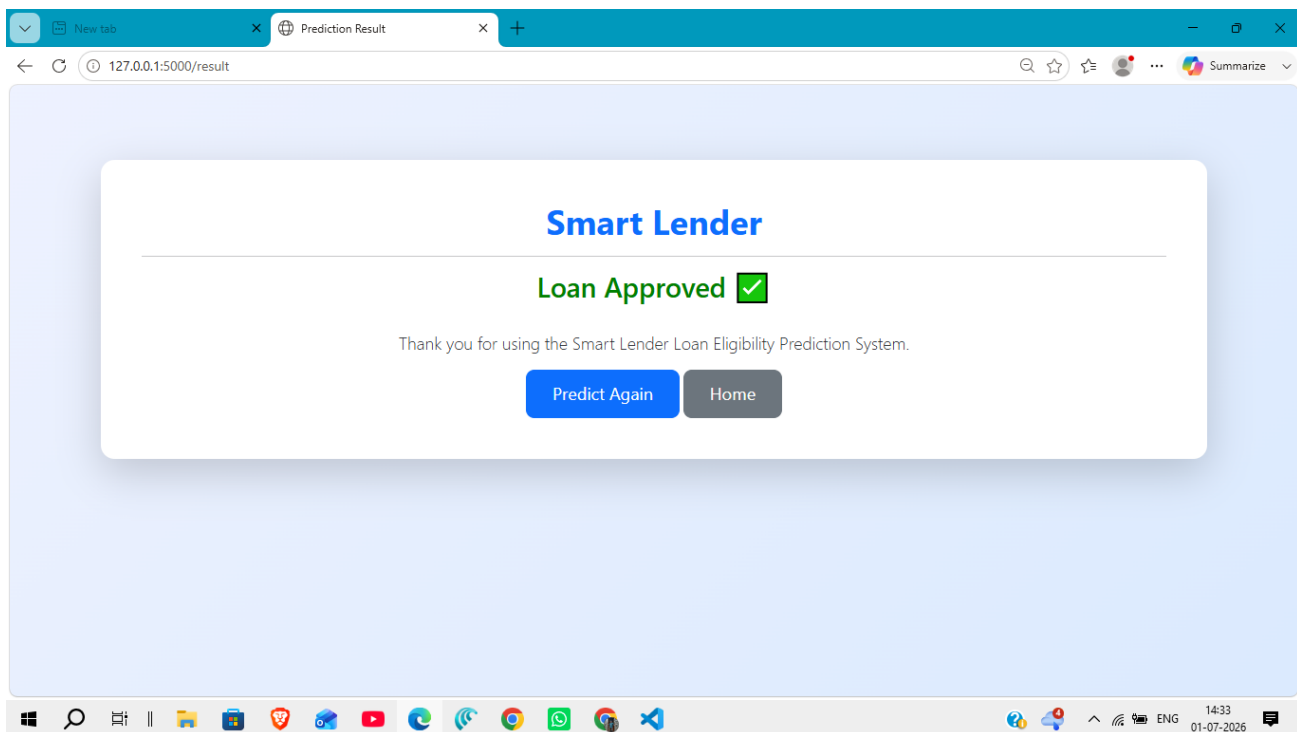
- Optimized data preprocessing before prediction.
- Loaded `model.pkl` and `scaler.pkl` only once during application startup.
- Reduced unnecessary preprocessing operations.
- Optimized the Flask prediction workflow for faster response.

Step 5: Screenshots / Evidence

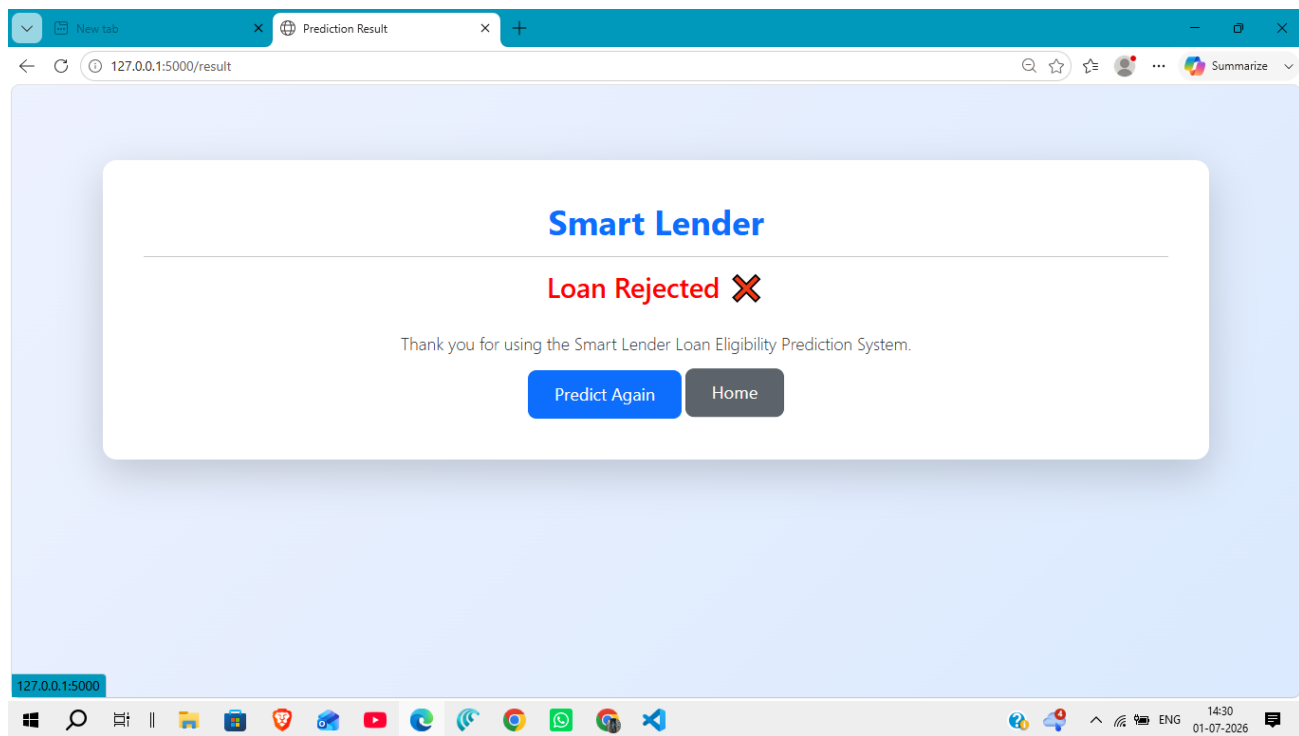
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Smart Lender Home Page



Loan Eligibility Prediction – Approved Result



Loan Eligibility Prediction – Rejected Result